

Ablation & Experiment Plan

*Seeing Seizures — Video vs. EEG Deep Learning for Rodent Seizure Detection & Grading
(WACV 2027)*

Prioritised to harden the paper’s claims against review. Tick items as runs complete.

Four load-bearing claims each ablation protects:

C1 video matches/beats EEG at *detection* **C2** detection is *backbone-agnostic* (4M ties 33M)

C3 video grades *gracefully* while EEG *collapses* **C4** video generalises more *stably* across unseen animals

C5 the detector attends to *behaviour*, not shortcuts (plus the “video-understanding” / “screening tool” framing).

Effort key: **Small** **Med** **Large** . Two expert reviews (claim→evidence gaps; peer-paper norms) converged on the Tier-1 four.

TIER 1 — Blocking. Reviewers will demand these.

☐ 1. Multi-seed runs + significance test **Med**

Defends: C1, C2. *Run:* ≥ 5 seeds for R(2+1)D and the EEG GRU on all three tasks; ≥ 5 seeds for the top/middle/bottom video backbones at detection; paired test (McNemar on labels, bootstrap on macro-F1). Report mean $\pm$ std in the results / EEG-arch / video-arch tables. *Risk if skipped:* “backbone-agnostic” asserts a 0.028 macro-F1 spread is noise, but with $n=1$ you cannot show it *is* noise. Seeds most likely *help* you. Also fixes the non-monotone data-efficiency dip (12 subj $\rightarrow$ 0.946).

☐ 2. Subject-disjoint CV for GRADING (3- and 5-class) **Med**

Defends: C3 — the paper’s most novel claim. *Run:* re-run 3/5-class grading (video + EEG) under the *same* subject-disjoint 5-fold splits already built for detection; report macro-F1 mean $\pm$ std and per-fold S4/S5 per-class F1. (Doing leave-one-subject-out, 20-fold, also strengthens C4’s “3.4 $\times$ more stable”). *Risk if skipped:* you call subject-disjoint “the honest test” yet give it only to the *weaker* claim (detection); grading uses the split you disown. **Caution:** 193 S5 clips are concentrated in a few animals, so holding them out may hurt — better you find out before a reviewer does.

☐ 3. Multi-channel EEG baseline **Med-Large**

Defends: C1, C3 (fairness of the cross-modal comparison). *Run:* train the best EEG models (GRU, TCN) on all available channels / a standard montage (not just the first channel) for detection + grading; report how much the gap narrows. **Free partial fix:** state that Ganguly — the EEG model you echo — is *also* single-channel, so you match the peer norm. *Risk if skipped:* the most credible “you hobbled the baseline” attack, and you already self-flag it. Currently no experimental rebuttal.

☐ 4. Single-frame (appearance-only) baseline **Small**

Defends: the “video-understanding problem” framing. *Run:* train a 2D image classifier (or the backbone with $T=1$) on one frame per clip, detection, same split; add the $T=1$ and $T=4$ points to the frames table. *Risk if skipped:* near-mandatory at a video venue. Your own frame ablation (0.960 $\rightarrow$ 0.970 $\rightarrow$ 0.972) hints a single frame may already reach ~ 0.95 — either you prove temporal dynamics matter, or you learn to soften the framing.

TIER 2 — Strong. Cheap credibility + preempts obvious questions.

☐ 5. Confusion matrices + ordinal metric for grading **Small** (do this even though it’s Tier 2 — cheapest big win)

Defends: “video degrades *gracefully*”. *Run:* report 3/5-class confusion matrices (video vs. EEG) and

an ordinal metric (quadratic-weighted κ / mean stage-distance). Predictions already exist. *Risk if skipped*: macro-F1 treats “S5→S4” the same as “S5→non-seizure,” so it cannot substantiate “graceful.” If video’s errors are off-by-one-stage while EEG’s scatter, that *proves* the story.

□ **6. Motion-representation input: RGB vs. optical-flow vs. frame-difference** Med

Defends: C5 + the core “seizure = whole-body motion” thesis; a *named* RodEpi ablation. *Run*: train the detector on each input type; a pure-motion input matching RGB shows the model keys on motion (complements Grad-CAM; echoes your RodEpi appearance-shortcut caution). *Risk if skipped*: RodEpi ablates exactly this; its absence is a visible gap vs. the benchmark you validate on.

□ **7. Operating-point / deployment metrics** Small

Defends: the “non-invasive screening tool / 3Rs / edge” framing. *Run*: specificity, precision at a fixed sensitivity, **false-alarms per hour**, PR curves, calibration (ECE) — mostly from existing scores. *Risk if skipped*: you cite vEpiNet’s one-third FP reduction but never report your own FP burden — the deployment metric for a screen.

□ **8. EEG window / stride / pooling + sampling-rate sweep** Small–Med

Defends: “Is the EEG baseline strong enough?”. *Run*: for the best EEG model sweep window (2/4/6/8s), stride, clip-aggregation (mean vs. attention pool), and decimation rate. (Prior analysis reportedly found 4s > 6s — worth confirming). *Risk if skipped*: you gave the *video* side a temporal ablation but the EEG side none; “architecture diversity ≠ tuning”.

□ **9. Class-imbalance-handling ablation** Med

Defends: the rare-stage grading numbers (S5 F1 0.42 vs. 0.03). *Run*: inverse-frequency weighting (current) vs. focal loss vs. resampling / class-balanced loss, both modalities, 5-class. *Risk if skipped*: one recipe is used; a reviewer asks whether EEG’s S5 collapse is *modality* or just a recipe better-matched to video.

□ **10. Pose / skeleton video baseline** Large (lighter stand-in: hand-crafted motion-energy baseline Small)

Defends: the “how far does *standard video* get” framing — the one method *family* you omit. *Run*: DeepLabCut/SLEAP keypoints → classifier (or a VSViG-style skeleton model), detection + grading. *Risk if skipped*: your two closest peers (Sabnis, VSViG) are *both* pose-based; either result is publishable, but omitting it is the most predictable community ask.

TIER 3 — Polish / robustness (several are cheap — infra already exists).

□ **11. Late-fusion headroom probe** Small

Defends: the “strongest system would fuse them” motivation. *Run*: one number: average-posterior or a logistic meta-learner over the video and EEG scores. Present as a probe, not a contribution — keeps “fusion out of scope” intact. *Risk if skipped*: preempts “why no fusion at all?”. (Already on your deferred list.).

□ **12. Temporal-shuffle / static-frame control** Small

Defends: that both your cohort *and* re-encoded RodEpi reflect genuine motion. *Run*: evaluate the trained detector with frames temporally shuffled; a drop ⇒ genuine dynamics, high accuracy ⇒ residual appearance leakage. *Risk if skipped*: you fixed the fps/frame-count/crop leak you *found*; a shuffle control is the standard proof none remains.

□ **13. Continuous sliding-window localisation** Med (RN197 frame-native infra reportedly exists)

Defends: the deployment / “screening” framing. *Run*: sliding-window inference over full sessions;

event-level precision/recall, false-alarms per hour, detection latency. *Risk if skipped:* all results assume pre-segmented clips; “what’s the false-alarm rate on raw hours of video?” is the deployment question.

□ **14. Cross-dataset zero-shot transfer (cohort $\leftrightarrow$ RodEpil)** Med

Defends: the generalisation contribution — goes *beyond* RodEpil. *Run:* train on your cohort, test on RodEpil (and reverse), zero-shot, with honest domain-shift caveats. *Risk if skipped:* your “external validation” currently *re-trains* on RodEpil (validates the method, not the trained model). A zero-shot number is a genuine differentiator.

□ **15. Measured real-time throughput** Small

Defends: “real-time, edge-side monitoring realistic” + the X3D efficiency headline. *Run:* FPS / ms-per-clip on named hardware for X3D vs. R(2+1)D. *Risk if skipped:* efficiency is argued only via parameter count; wall-clock substantiates it cheaply.

□ **16. Background / crop robustness** Small-Med

Defends: C5 (“behaviour, not cage background”) — a quantitative Grad-CAM. *Run:* background-masking (keep only the animal) or train-on-one-crop / test-on-another; show accuracy is preserved. *Risk if skipped:* Grad-CAM is only qualitative; this rules out cage-framing shortcuts under same-subject session splits.

□ **17. Inter-annotator agreement / human ceiling** Med (human cost)

Defends: the interpretability of all grading numbers. *Run:* a second expert re-grades a stratified subset (~300–500 clips); report weighted Cohen’s κ as a noise ceiling. *Risk if skipped:* without a ceiling, 0.66 macro-F1 is uninterpretable — it may already be at human agreement (i.e. video is near-optimal).

Free wins — text only, no experiments (I can apply these now).

□ **Ganguly single-channel defence** in Sec. 3 — neutralises the EEG-fairness objection (#3) for free.

□ **Sec. 7 caveats:** grading is validated on a *single* cohort (RodEpil is binary-only), and a *single* annotator graded (label-noise limitation).

If you only do a handful (both reviews converged here): **#2** subject-disjoint grading, **#1** seeds, **#5** confusion/ordinal, **#3** multi-channel EEG (or at least the free Ganguly text defence), plus **#4** single-frame (tiny, and it’s a video venue). That set hardens all four load-bearing claims.

Leverages you already have: subject-disjoint fold infra $\rightarrow$ #2; existing predictions $\rightarrow$ #5, #7; RN197 sliding-window $\rightarrow$ #13; the RodEpil pipeline $\rightarrow$ #14.